

1. Success Criteria (Acceptance Tests)

The prototype will be considered successful if it meets the following criteria during validation:

Category	Metric	Target
Mechanical	Pan range	180° minimum continuous
	Tilt range	-50° to +30°
	Load capacity	Holds laser module and any counterweights without sag
Electrical	Power consumption	≤ 15 W average at 12 V
	Battery backup	Can run 2 hours on a small LiFePO ₄ battery (if solar not used)
Control	Positioning accuracy	±2° (static)
	Response time (0-60° step)	< 0.3 seconds

	Smoothness	No visible jitter or overshoot
Safety	Laser cut-off	Relay de-energises laser in < 10 ms on emergency stop
	Geofencing	Software prevents laser pointing outside defined area (simulated)
Tracking (basic)	Target following	Can track a coloured ball moving at ~0.5 m/s at 5 m distance for 30 s
Environmental	Operating temperature	0°C to 45°C (simulate)
	Dust/water resistance	Enclosure IP54 (test with simulated dust/water spray)

2. Documentation Deliverables (End of Project)

1. **User Manual:** How to power up, operate, and troubleshoot the turret.
2. **Technical Reference:** Schematics, wiring diagrams, bill of materials, datasheets.
3. **Software Repository:** Well-commented source code, build instructions, dependency list.
4. **Test Report:** Detailed results of all validation tests, including raw data and analysis.
5. **Demonstration Video:** 3-5 minute video showing the turret in action (pointing, tracking, safety features).

6. **Lessons Learned:** What worked, what didn't, recommendations for the full BirdGuard system.